

Quantum Mechanics cheat sheet

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References

- [1] János A. Bergou and Mark Hillery, Introduction to the Theory of Quantum Information Processing, Springer 2015
- [2] N. David Mermin, Quantum mysteries revisited, American Journal of Physics 58, 731 1990
- [3] Kurt Jacobs, Quantum Measurement Theory, Cambridge 2014

Classical probability	
(P) Bayes theorem	Bayes theorem states that the posterior probability for a random variable X after obtaining the value y for a random variable Y is obtained by multiplying the prior on X by the likelihood function $P(Y = y X = x)$ and by normalizing the result such that the sum over x gives $P(Y = y)$: $P(X = x Y = y) = \frac{P(X=x)P(Y=y X=x)}{\mathcal{N}}$, where $\mathcal{N} = \int_{-\infty}^{\infty} P(Y = y X = x)P(X = x)dx = P(Y = y)$

Some standard formulae	
(P) Complex conjugate	(1) $\overline{z + w} = \overline{z} + \overline{w}$ (2) $\overline{z \times w} = \overline{z} \times \overline{w}$ (3) $z ^2 = z\overline{z}$ (4) $z^{-1} = \frac{\overline{z}}{ z ^2}$ if $z \neq 0$, $\frac{1}{a+ib} = \frac{a-ib}{a^2+b^2}$ for a, b non-zero real (5) $\exp(\overline{z}) = \overline{\exp(z)}$ (6) $\ln(\overline{z}) = \overline{\ln(z)}$ if $z \neq 0$
(P) Trigonometry	(1) $\cos(a + b) = \cos a \cos b - \sin a \sin b$ (2) $\sin(a + b) = \sin a \cos b + \cos a \sin b$ (3) $\cos a \cos b = \frac{1}{2} [\cos(a - b) + \cos(a + b)]$ (4) $\sin a \sin b = \frac{1}{2} [\cos(a - b) - \cos(a + b)]$ (5) $\sin a \cos b = \frac{1}{2} [\sin(a - b) + \sin(a + b)]$ (6) $\cos p + \cos q = 2 \cos \frac{p+q}{2} \cos \frac{p-q}{2}$ (7) $\sin p + \sin q = 2 \sin \frac{p+q}{2} \cos \frac{p-q}{2}$ (8) $\cos x = \frac{e^{ix} + e^{-ix}}{2}$, $\sin x = \frac{e^{ix} - e^{-ix}}{2i}$ (9) $\sin \frac{\pi}{6} = 1/2$, $\sin \frac{\pi}{4} = \sqrt{2}/2$, $\sin \frac{\pi}{3} = \sqrt{3}/2$
(P) Dirac Distribution	(1) $\int_{-\infty}^{\infty} f(x)\delta(x-a)dx = f(a)$ for any function $f : \mathbf{R} \rightarrow \mathbf{R}$ (2) $\delta(x-a) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{ip(x-a)} dp$ (3) $\delta(\alpha x) = \frac{\delta(x)}{ \alpha }$ (4) $\delta(a-x) = \delta(x-a)$ (5) $\delta(g(x)) = \sum_i \frac{\delta(x-x_i)}{ g'(x_i) }$ where x_i are roots of function g (6) $\int_{-\infty}^{\infty} f(x)\delta^{(k)}(x-a)dx = (-1)^k f^{(k)}(a)$ (7) $\int_{-\infty}^{\infty} f(x)\delta(x-a)dx = \int_{-\infty}^{\infty} f(x)H'(x-a)dx$ (8) $\frac{1}{\sigma\sqrt{\pi}} \exp\left[-\left(\frac{x-a}{\sigma}\right)^2\right] \rightarrow \delta(x-a)$ for $\sigma \rightarrow 0$
(P) Fourier Transform for function $f : \mathbf{R} \rightarrow \mathbf{C}$	(1) $f(x) = \int_{-\infty}^{\infty} \widehat{f}(p)e^{i2\pi xp}dp$ with $\widehat{f}(p) = \int_{-\infty}^{\infty} f(x)e^{-i2\pi xp}dx$ (2) $\int_{-\infty}^{\infty} f(x)\overline{g(x)}dx = \int_{-\infty}^{\infty} \widehat{f}(p)\widehat{\overline{g}}(p)dp$ (Parseval) (3) if $h(x) = (f * g)(x) = \int_{-\infty}^{\infty} f(y)g(x-y)dy$ then $\widehat{h}(p) = \widehat{f}(p) \cdot \widehat{g}(p)$ (convolution)
(P) Fourier Series for periodic function $f : [0, 2\pi] \rightarrow \mathbf{C}$	(1) $f(x) = \sum_{n=-\infty}^{\infty} \widehat{f}_n e^{i2\pi xn}$ with $\widehat{f}_n = \int_0^{2\pi} f(x)e^{-i2\pi xn}dx$ (2) $\int_{-\infty}^{\infty} f(x)g(x)dx = \sum_{n=-\infty}^{\infty} \widehat{f}_n \widehat{\overline{g}}_n$ (Parseval)
(P) Stirling approximation	$N! \approx N^N e^{-N} \sqrt{2\pi N}$ $\ln(N!) \approx N \ln N - N$

Matrix theory	
(Note) We denote by $M_n(\mathbf{C})$ the set of n -dimensional square matrices with entries in $\mathbf{C}$.	
(D) Unitary matrix	$U \in M_n(\mathbf{C})$ is unitary $\iff U^\dagger = U^{-1}$
(P) Properties for unitary matrices	U is unitary $\Rightarrow$ (1) U can be diagonalized by a unitary matrix (2) all eigenvalues of U have norm 1 (3) the columns of U form an orthonormal basis of $\mathbf{C}^n$ (4) the rows of U form an orthonormal basis of $\mathbf{C}^n$ (5) there exists a hermitian matrix H such that $U = \exp[iH]$
(D) Hermitian matrix	$A \in M_n(\mathbf{C})$ is hermitian $\iff A^\dagger = A$
(P) Properties for hermitian matrices	A is hermitian $\Rightarrow$ (1) A can be diagonalized by a unitary matrix (2) all eigenvalues of A are real numbers
(P) Polar decomposition	Any matrix $A \in M_n(\mathbf{C})$ can be decomposed in the form $A = UP = QU$ where U is unitary, and $P = \sqrt{A^\dagger A}$ and $Q = \sqrt{AA^\dagger}$ are positive semidefinite (i.e. for all $x \in \mathbf{C}^n$, $x^\dagger Px \geq 0$), hermitian matrices ($A^\dagger A$ is non negative because $\forall x, \ Ax\ ^2 = x^\dagger A^\dagger Ax \geq 0$).
(D) Exponential of a matrix	For any matrix $A \in M_n(\mathbf{C})$, $\exp(A) = e^A = \sum_{k=1}^{\infty} \frac{A^k}{k!}$ The above series always converges.
(P) Properties	(1) if $[X, Y] = XY - YX = 0$ then $e^{X+Y} = e^X e^Y$ (2) if $[X, Y]$ commutes with X and Y, $e^X e^Y = e^{X+Y+\frac{1}{2}[X,Y]}$ (3) otherwise need to apply Baker-Campbell-Hausdorff formula (4) if Y is invertible, $e^{YXY^{-1}} = Y e^X Y^{-1}$
(D) Logarithm of a matrix	For any matrix $A \in M_n(\mathbf{C})$, the matrix $B \in M_n(\mathbf{C})$ is a logarithm of A if $\exp(B) = A$. It is usually not unique.
(P) Property	For any matrix $A \in M_n(\mathbf{C})$, and any unitary matrix $U \in M_n(\mathbf{C})$, $\exp(U^\dagger A U) = U^\dagger \exp(A) U$, $\ln(U^\dagger A U) = U^\dagger \ln(A) U$, provided $\ln(A)$ exists.

Basics linear algebra	
(Note) For any Hilbert space H , the set of linear maps on H is denoted $\mathcal{L}(H)$.	
(D) Trace	Let H_n be a n -dimensional Hilbert space and let $X \in \mathcal{L}(H_n)$. The trace of X is defined as $\text{Tr}(X) = \sum_i \langle e_i X e_i \rangle$ where $\{ e_i\rangle\}_{i=1,\dots,n}$ is any orthonormal basis of H_n .
(P) Properties	(1) the above definition is independent of the choice of basis, (2) for any operators X and Y on H_n , $\text{Tr}(XY) = \text{Tr}(YX)$
(Note) Let $H_A \otimes H_B$ be the product space of the Hilbert spaces H_A and H_B endowed with the respective orthonormal bases $\{ e_a\rangle_A\}_{a=1,2,\dots}$ and $\{ e_b\rangle_B\}_{b=1,2,\dots}$.	
(D) Partial inner product	Let $ \phi\rangle_A$ be a state in H_A . The partial inner product $\langle\phi _A$ is defined as $\langle\phi _A : H_A \otimes H_B \rightarrow H_B$ such that, for any state $ \psi\rangle = \sum_{a,b} \mu_{ab} e_a\rangle_A \otimes e_b\rangle_B \in H_A \otimes H_B$, $\langle\phi _A \psi\rangle = \sum_{a,b} \mu_{ab} \langle\phi _A e_a\rangle_A \otimes e_b\rangle_B$ We also adopt the notation $\langle\psi \phi\rangle_A = (\langle\phi _A \psi\rangle)^\dagger = \sum_{a,b} \mu_{ab}^* \langle e _A \phi\rangle_A \otimes \langle e_b _B$
(D) Partial trace	The partial trace over H_A , Tr_A is defined as: $\text{Tr}_A : \mathcal{L}(H_A \otimes H_B) \rightarrow \mathcal{L}(H_B)$ where, for any linear map $X \in \mathcal{L}(H_A \otimes H_B)$, $\text{Tr}_A(X) = \sum_a \langle e_a _A X e_a\rangle_A$

Density matrix	
(Note) Let H_n the Hilbert space of dimension n .	
(D) Density matrix	The matrix $\rho \in \mathcal{L}(H_n)$ is a density matrix if there exists $\left[\begin{array}{l} \text{a set } \{p_i\}_{i=1,2,\dots}, \text{ with } \sum_i p_i = 1 \text{ and } p_i \in [0, 1] \text{ for all } i, \text{ and,} \\ \{ \phi_i\rangle\}_{i=1,2,\dots} \text{ with } \phi_i\rangle \in H_n \end{array} \right]$ such that $\rho = \sum_i p_i \phi_i\rangle \langle \phi_i .$ We say that such ρ describes a pure-state ensemble in which the system is in state $\phi_i\rangle$ with probability p_i, for $i = 1, 2, \dots$
(D) Special types of density matrices	(pure) ρ is pure if there exists $\phi\rangle \in H_n$ such that $\rho = \phi\rangle \langle \phi$ (mixed) ρ is mixed if it is not pure (completely mixed) ρ is completely mixed if $\rho = \frac{1}{n} \mathbf{1}$
(P) Equivalent definition	An operator ρ is a density matrix $\iff$ $\left[\begin{array}{l} \text{(1) } \rho \text{ is hermitian,} \\ \text{(2) } \text{Tr}(\rho) = 1, \\ \text{(3) } \rho \text{ is positive, i.e. } \forall \psi\rangle \in H_n, \langle \psi \rho \psi \rangle \geq 0, \end{array} \right]$
(P) Pure state	$\rho \text{ is pure } \iff \text{Tr}(\rho^2) = 1$
(P) Ensemble decompositions	A density matrix can be written as $\sum_{i=1}^A \alpha_i a_i\rangle \langle a_i$ and $\sum_{j=1}^B \beta_j b_j\rangle \langle b_j$ with $A \geq B \iff$ There exists a unitary matrix U of dimension A such that: $\sqrt{\alpha_i} a_i\rangle = \sum_{j=1}^B U_{ij} \sqrt{\beta_j} b_j\rangle.$
(P) Schmidt decomposition	Let $H = H_A \otimes H_B$ be the product space of the Hilbert spaces H_A and H_B. For any state $\Psi_{AB}\rangle \in H$, there exist orthonormal bases $\{ a_i\rangle\}_i$ for H_A and $\{ b_j\rangle\}_j$ for H_B such that: $ \Psi_{AB}\rangle = \sum_{k=1}^n \sqrt{\lambda_k} a_k\rangle \otimes b_k\rangle$ where $n \leq \min(\dim(H_A), \dim(H_B))$. Note that the partial traces $\text{Tr}_A(\Psi_{AB}\rangle \langle \Psi_{AB})$ and $\text{Tr}_B(\Psi_{AB}\rangle \langle \Psi_{AB})$ share the same non-zero eigenvalues.
(Proof) Let $\{ a_i\rangle\}_i$ be the basis diagonalizing the partial density $\rho_A = \text{Tr}_B(\Psi_{AB}\rangle \langle \Psi_{AB})$. For some basis $\{ v_j\rangle\}_j$ of H_B, the state $\Psi_{AB}\rangle$ can be written as $\Psi_{AB}\rangle = \sum_{i,j} c_{i,j} a_i\rangle \otimes v_j\rangle = \sum_i a_i\rangle \otimes \tilde{b}_i\rangle$ where $\tilde{b}_i\rangle = \sum_j c_{i,j} v_j\rangle$. We then have $\rho_A = \sum_{i,j} \langle \tilde{b}_j \tilde{b}_i \rangle a_i\rangle \langle a_j$ thus $\langle \tilde{b}_j \tilde{b}_i \rangle = \delta_{i,j} \lambda_i$ where λ_i are the eigenvalues of ρ_A. Normalizing the states $\tilde{b}_i\rangle$ we get the result.	
(P) Purification	For any density matrix $\rho_A = \sum_i p_i \psi_i\rangle_A \langle \psi_i _A$ where $\psi_i\rangle_A$ are any states in H_A, there exists a Hilbert space H_B and an orthonormal basis $\{ b_j\rangle\}_j$ for H_B such that $\rho_A = \text{Tr}_B(\Psi_{AB}\rangle \langle \Psi_{AB}), \quad \text{where}$ $ \Psi_{AB}\rangle = \sum_i \sqrt{p_i} \psi_i\rangle_A \otimes b_i\rangle.$
(D) Placing	When the partial trace of a universe state, $\psi\rangle$, gives the density state ρ for a subsystem, we say that $\psi\rangle$ places the system in the state ρ.

Bloch sphere	
(D) Pauli Matrices	$\sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \quad \sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \quad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$
(D) one qubit representation	A density operator for a single qubit can be expressed as $\rho = \frac{1}{2}(\mathbf{1} + \vec{u} \cdot \vec{\sigma})$ where $\vec{u}$ is a vector of norm less or equal to unity. When $\ \vec{u}\ = 1$ the state ρ is pure, when $\ \vec{u}\ = 0$ it is completely mixed.

entanglement	
(D) for pure states	A pure state $ \psi\rangle_{AB}$ in product space $H_A \otimes H_B$ is not entangled if it can be written as a product of two pure states in H_A and H_B , i.e. $ \psi\rangle_{AB} = \psi\rangle_A \otimes \psi\rangle_B$. Otherwise it is entangled .
(D) for density matrices	A density matrix ρ_{AB} in $\mathcal{L}(H_A \otimes H_B)$ is separable if it can be written as a mixture of product of states, i.e. $\rho = \sum_i p_i \rho_{Ai} \otimes \rho_{Bi}$ for some $p_i \in [0, 1]$, $\rho_{Ai} \in \mathcal{L}(H_A)$, $\rho_{Bi} \in \mathcal{L}(H_B)$. Otherwise it is entangled .
(D) maximally entangled	A state is maximally entangled if it is entangled and its reduced density matrices are proportional to identity.
(P) classical correlation	For a pure state that is not entangled, measurements on systems A and B are not correlated. For a separable density matrix, measurements on systems A and B are only classically correlated, or not correlated.
(P) Bell or CSHS inequality	Let Alice and Bob be two remote experimenters. Alice makes a measurement a or a' , both measurements can each give $+1$ or -1 . Independently of Alice, Bob makes the measurement b or b' , both measurement can each give $+1$ or -1 . Then, if there exists a hidden variable λ that gives the measurement result for all combinations of the measurement choice, then we have the Clauser Horne Shimony Holt inequality: $-2 \leq E[a \times b] + E[a \times b'] + E[a' \times b] - E[a' \times b'] \leq 2$
(Proof) $E[a \times b] + E[a \times b'] + E[a' \times b] - E[a' \times b'] = \int P(\lambda) d\lambda [a(\lambda)(b(\lambda) + b'(\lambda)) + a'(\lambda)(b(\lambda) - b'(\lambda))] \leq 2$ because $b \pm b'$ is 0 or 2 (one is 0 when the other one is 2).	
(P) violation of Bell inequality	Let a be the measurement along σ_x and a' be along σ_y and similarly for b and b' . Let the initial state be $ \Psi\rangle = \frac{1}{\sqrt{2}}(00\rangle + e^{i\pi/4} 11\rangle)$. Then we get the violation of the above inequality with the term in the middle equal $2\sqrt{2}$.

(P) Mermin-GHZ paradox	Let A, B and C, three remote experimenters perform the measurement of the observable x or y (exclusive or), the measurement yielding $+1$ or -1 with both observables. Assume that the setup is so that the product of A, B, C's measurement outcomes is $+1$ in case A, B, C chose the bases x, y, y or y, x, y or y, y, x respectively. Then a classical theory based on hidden variables predicts that such product must be $+1$ for the bases x, x, x. Quantum mechanics violates this classical prediction with the state $\Psi\rangle = \frac{1}{\sqrt{2}}(+++\rangle - --\rangle)$ and the observables $x = \sigma_x$, $y = \sigma_y$ and the states $+\rangle$ and $-\rangle$ are the eigenstates of σ_z with eigenvalues $+1$ and -1 respectively, with σ_i being the Pauli matrices.																																																																																						
(Proof) A hidden variable theory would tell independently to each detector the result of the measurement of the observation depending on the choice of the observable of the experimenter (so the outcome given the choice of x or y for each experimenter. This table must be separate for each experimenter as they are remote). This would look like <table><tr><td></td><td>A</td><td>B</td><td>C</td><td></td></tr><tr><td>x</td><td>$+$</td><td>$-$</td><td>$-$</td><td rowspan="2">meaning A, B, C will observe -1, -1, $+1$ if they chose</td></tr><tr><td>y</td><td>$-$</td><td>$+$</td><td>$+$</td></tr></table> y, x, y, giving the product $+1$. There are $2^6 = 64$ binary tables of size 6, but you have the conditions $X_A Y_B Y_C = 1$, $Y_A X_B Y_C = 1$, $Y_A Y_B X_C = 1$. Multiply the first equation by $Y_B Y_C$, the second by $Y_A Y_C$ and the third by $Y_A Y_B$ and you get (as $Y_i^2 = 1$) $X_A = Y_B Y_C$, $X_B = Y_A Y_C$, $X_C = Y_A Y_B$. You can set -1 or $+1$ for each Y_i which gives the 8 possible tables <table><tr><td></td><td>A</td><td>B</td><td>C</td><td></td><td></td><td></td><td></td></tr><tr><td>x</td><td>$+$</td><td>$+$</td><td>$+$</td><td>x</td><td>$+$</td><td>$-$</td><td>$-$</td></tr><tr><td>y</td><td>$+$</td><td>$+$</td><td>$+$</td><td>y</td><td>$+$</td><td>$-$</td><td>$-$</td></tr><tr><td></td><td>A</td><td>B</td><td>C</td><td></td><td>A</td><td>B</td><td>C</td></tr><tr><td>x</td><td>$+$</td><td>$-$</td><td>$-$</td><td>x</td><td>$-$</td><td>$+$</td><td>$-$</td></tr><tr><td>y</td><td>$-$</td><td>$+$</td><td>$+$</td><td>y</td><td>$-$</td><td>$+$</td><td>$-$</td></tr><tr><td></td><td>A</td><td>B</td><td>C</td><td></td><td>A</td><td>B</td><td>C</td></tr><tr><td>x</td><td>$+$</td><td>$+$</td><td>$+$</td><td>x</td><td>$-$</td><td>$-$</td><td>$+$</td></tr><tr><td>y</td><td>$-$</td><td>$-$</td><td>$-$</td><td>y</td><td>$+$</td><td>$+$</td><td>$-$</td></tr></table> From this you get that for the choice x, x, x the product of the outcomes is always $+1$. The violation with the quantum setup is straightforward to check, given that $\sigma_x \pm\rangle = \mp\rangle$ and $\sigma_y \pm\rangle = \pm i \mp\rangle$			A	B	C		x	$+$	$-$	$-$	meaning A , B , C will observe -1 , -1 , $+1$ if they chose	y	$-$	$+$	$+$		A	B	C					x	$+$	$+$	$+$	x	$+$	$-$	$-$	y	$+$	$+$	$+$	y	$+$	$-$	$-$		A	B	C		A	B	C	x	$+$	$-$	$-$	x	$-$	$+$	$-$	y	$-$	$+$	$+$	y	$-$	$+$	$-$		A	B	C		A	B	C	x	$+$	$+$	$+$	x	$-$	$-$	$+$	y	$-$	$-$	$-$	y	$+$	$+$	$-$
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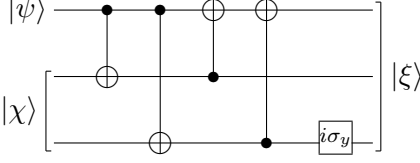
Measurement		
(D)	von Neumann measurement	Let $\rho \in \mathcal{L}(H)$ be the density matrix giving the state of the system prior to the measurement. The von Neumann measurement in the orthonormal basis $\{ i\rangle\}_i$ of H finds the system in the state $\tilde{\rho}_i = \frac{P_i \rho P_i}{\text{Tr}(P_i \rho P_i)}$ with the probability $p_i = \text{Tr}(P_i \rho P_i)$ where $P_i = i\rangle \langle i$ is the projection operator onto the basis state $i\rangle$. In this case, note that $\tilde{\rho}_i$ is simply equal to P_i. The state of the system after the measurement can be described in the mixed form: $\sum_i p_i \tilde{\rho}_i = \sum_i P_i \rho P_i,$ not forgetting that the result of the measurement is stored somewhere outside the system, keeping the evolution of the system plus the outer world unitary.
(D)	Generalized measurement	Let $\rho \in \mathcal{L}(H)$ be the density matrix giving the state of the system prior to the measurement. Any set of operators of $\mathcal{L}(H)$, $\{A_i\}_i$ such that $\sum_i A_i^\dagger A_i = \mathbf{1}$ describes a possible measurement on the system, where the measurement finds the system in state $\tilde{\rho}_i = \frac{A_i \rho A_i^\dagger}{\text{Tr}(A_i \rho A_i^\dagger)}$ with the probability $p_i = \text{Tr}(A_i \rho A_i^\dagger).$ Note that the operators A_i are not necessarily hermitian nor unitary. Again, the state of the system restricted to H after the measurement can be described as: $\sum_i p_i \tilde{\rho}_i = \sum_i A_i \rho A_i^\dagger$
(P)	Interpreting generalized measurement	Applying the polar decomposition $A_i = U_i P_i$ for each A_i that form the generalized measurement $\{A_i\}_i$, we have that: $\sum_i A_i^\dagger A_i = \sum_i (P_i)^2 = \mathbf{1}$ and the measurement finds the system in state: $\tilde{\rho}_i = U_i \left(\frac{P_i \rho P_i}{p_i} \right) U_i^\dagger$ with the probability $p_i = \text{Tr}(P_i \rho P_i) = \text{Tr}(P_i^2 \rho),$ where the P_i are positive semidefinite hermitian operators and U_i are unitary. As unitary transformation does not change the eigenvalues and can be interpreted as time evolution, the above formula describes an information gathering by the operator P_i followed by a feedback in which we apply U_i depending on the result i of the information gathering measurement.

(P) Generalized measurement is a von Neumann measurement	Any generalized measurement on a Hilbert space H can be described as a von Neumann measurement on a larger Hilbert space $H \otimes H_E$. More precisely, let $\{A_i\}_{i=1,\dots,n}$ describe the generalized measurement on H. Then there exist a Hilbert space H_E of dimension n, a unitary operator $U \in \mathcal{L}(H_E \otimes H)$ and an orthonormal basis of H_E, $\{ i\rangle_E\}_{i=1,\dots,n}$ so that, for each $i = 1, \dots, n$, $\forall \rho \in \mathcal{L}(H),$ $A_i \rho A_i^\dagger = \text{Tr}_E \left((P_i \otimes \mathbf{1}_H) U \{ 1\rangle_E \langle 1 _E \otimes \rho\} U^\dagger (P_i \otimes \mathbf{1}_H) \right)$ where we denote $P_i = i\rangle_E \langle i _E$ the projection operators on H_E. Usually, the space H is called the target and the space H_E the probe space.
(Proof) Let h be the dimension of the space H. Consider the matrix of width h and height $n \times h$ formed the matrices A_i stacked in column. This matrix defines h vectors $\{\vec{u}_j\}_j$, of $H_E \otimes H$ that are orthonormal between themselves, because $\sum_i A_i^\dagger A_i = \mathbf{1}$. Indeed, $\vec{u}_j \cdot \vec{u}_k = \sum_i \sum_l (A_i)_{l,j}^* (A_i)_{l,k} = \sum_i \sum_l (A_i^\dagger)_{j,l} (A_i)_{l,k} = \sum_i (A_i^\dagger A_i)_{j,k} = \delta_{j,k}$. Therefore by completing the orthonormal basis (for the remaining $n \times h - h$ vectors), there exists a unitary matrix U of dimension $n \times h$ such that its first h columns are given by the stacked matrices A_i. That is, $U = \sum_{i,j=1,\dots,n} i\rangle_E \langle j _E \otimes B_{i,j}$ where the $B_{i,j}$ are h dimensional and $B_{i,1} = A_i$ for all i. Now suppose that the probe space is initially in state $1\rangle_E$. The initial combined state is $1\rangle_E \langle 1 _E \otimes \rho$ and by applying the unitary operator U and (partially) projecting with P_i on the probe space, we get: $(P_i \otimes \mathbf{1}_H) U \{ 1\rangle_E \langle 1 _E \otimes \rho\} U^\dagger (P_i \otimes \mathbf{1}_H) = (P_i \otimes \mathbf{1}_H) i\rangle_E \langle 1 _E \otimes B_{i,1} \{ 1\rangle_E \langle 1 _E \otimes \rho\} 1\rangle_E \langle i _E \otimes B_{i,1}^\dagger$ and tracing over H_E we obtain, using $B_{i,1} = A_i$, $\text{Tr}_E ((P_i \otimes \mathbf{1}_H) U \{ 1\rangle_E \langle 1 _E \otimes \rho\} U^\dagger (P_i \otimes \mathbf{1}_H)) = A_i \rho A_i^\dagger$.	(P) Sequence of measurement as a single measurement All processes that involves sequences of measurements with intermediary evolution depending on the results of past measurements, can be described as a single unitary process followed by a single measurement performed at the end.
(Proof) Let $\{A_i\}_{i=1,\dots,n}$ be a general measurement and suppose that we apply the unitary transformation U_i upon seeing the result i. Then $\{U_i A_i\}_{i=1,\dots,n}$ is also a general measurement as $A_i^\dagger U_i^\dagger U_i A_i = A_i^\dagger A_i$, and therefore there exists H_E, $U \in \mathcal{L}(H_E \otimes H)$ and projection operators $P_i \in \mathcal{L}(H_E)$ such that: $U_i A_i \rho A_i^\dagger U_i^\dagger = \text{Tr}_E ((P_i \otimes \mathbf{1}_H) U \{ 1\rangle_E \langle 1 _E \otimes \rho\} U^\dagger (P_i \otimes \mathbf{1}_H))$. The density operator after this first measurement and feedback is $\sum_i U_i A_i \rho A_i^\dagger U_i^\dagger = \sum_i \text{Tr}_E ((P_i \otimes \mathbf{1}_H) U \{ 1\rangle_E \langle 1 _E \otimes \rho\} U^\dagger (P_i \otimes \mathbf{1}_H))$ $= \sum_i \text{Tr}_E ((P_i \otimes \mathbf{1}_H) U \{ 1\rangle_E \langle 1 _E \otimes \rho\} U^\dagger) = \text{Tr}_E (U \{ 1\rangle_E \langle 1 _E \otimes \rho\} U^\dagger)$ because P_i are projectors. Now you can reiterate for the second general measurement that can depend on the first outcome i, $\{B_{i,j}\}_{i,j}$ followed by the unitary transformations $V_{i,j}$, to get the density matrix after the measurement as $\sum_{i,j} V_{i,j} B_{i,j} U_i A_i \rho A_i^\dagger U_i^\dagger B_{i,j}^\dagger V_{i,j}^\dagger = \text{Tr}_{E \otimes F} (V U \{ 1, 1\rangle_{E,F} \langle 1, 1 _{E,F} \otimes \rho\} U^\dagger V^\dagger)$ which can be read as a unitary transformation on $H_F \otimes H_E \otimes H$ followed by a final projection on $H_E \otimes H_F$ with a pre-defined projectors $P_i \otimes P_j$ (you cannot change the projectors as they are conditioned by the iterated measurements).	

von Neumann entropy	
(D) von Neumann entropy	For any density matrix $\rho \in \mathcal{L}(H_n)$, the von Neumann entropy of state ρ is defined as $S(\rho) = -\text{Tr}(\rho \ln \rho) = -\sum_{i=1}^n \lambda_i \ln \lambda_i$ where the $\{\lambda_i\}_{i=1,\dots,n}$ are the eigenvalues of ρ.
(P) Properties	(1) $0 \leq S(\rho) \leq \ln(n)$ (2) $S(\rho) = 0$ if and only if ρ is pure (3) $S(\rho) = \ln(n)$ if and only if ρ is completely mixed (4) S is invariant under unitary transform, i.e. $S(U\rho U^\dagger) = S(\rho)$ (concavity) for ρ_k density matrices and p_k positive, $\sum_k p_k = 1$ $S(\sum_k p_k \rho_k) \geq \sum_k p_k S(\rho_k)$
(P) Subadditivity	If ρ is a density for $H_A \otimes H_B$, and $\rho_A = \text{Tr}_B(\rho)$, $\rho_B = \text{Tr}_A(\rho)$ are the densities for H_A and H_B, then (1) $S(\rho) \leq S(\rho_A) + S(\rho_B)$ with equality iff $\rho = \rho_A \otimes \rho_B$ (2) $S(\rho_A) - S(\rho_B) \leq S(\rho)$
(P) Strong subadditivity	for any three systems A, B, and C, $S(\rho_{ABC}) + S(\rho_B) \leq S(\rho_{AB}) + S(\rho_{BC}).$

thermodynamic entropy	
(D) macro and micro states	The macrostate of a system is defined by those global physical properties that one can measure with simple macroscopic apparatus. The mutually orthogonal quantum states that the system could be in, given its macrostate, are the accessible microstates of the system for this macrostate. The observer has access only to the macrostate.
(D) thermodynamic entropy	The thermodynamic entropy of a system associated with a given macrostate, is defined as the logarithm of the total number Ω of accessible microstates for this macrostate (possibly multiplied by the Boltzmann's constant $k = 1.38064 \times 10^{-23}$) $S = k \ln(\Omega)$
(P) Fundamental postulate of statistical mechanics	All the accessible microstates of a closed system are equally likely. Remark: if we have two systems that are interacting, the microstates of the combined system will be equally likely, but this may not be true for those of the subsystems.
(P) Possible explanation of the fundamental postulate	Denote by ρ the density state of a system that would have been obtained by tracing out the rest of the universe, starting from a completely mixed state for the universe. It turns out that almost all pure states for the universe would place the system in a state that is very close to ρ , i.e. tracing out the complement of the system yields a density state close to ρ for the system.
(P) Second law of thermodynamics	The fundamental postulate of statistical mechanics implies that a system will always evolve to the macrostate with the highest entropy (i.e. the maximum number of microstates). For instance, a macrostate in which particles of gas are evenly spread throughout a room has way more microstates than a macrostate in which these particles are confined in a small volume of the room. The origin of irreversibility is not a time-asymmetry in the evolution of quantum mechanics, but the asymmetry between the macrostates, in that some have more microstates than others. If we start a system in a state that is out of equilibrium and run the evolution backward in time, the system will equilibrate in the same way as it equilibrates forward in time.
(P) Landauer's erasure principle	When a single bit of information is erased from a system (i.e. we take a qubit whose initial state is $ 0\rangle$ or $ 1\rangle$ and apply a process which transforms the state into $ 0\rangle$), the entropy of the environment must increase by $k \ln(2)$.

quantum eraser	
(Note)	$\boxed{H} = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \quad \boxed{\phi} = \begin{pmatrix} 1 & 0 \\ 0 & \exp(i\phi) \end{pmatrix}.$
(P) Step 1: simple interferometer	The following setup $ 0\rangle \text{---} \boxed{H} \text{---} \boxed{\phi} \text{---} \boxed{H} \text{---}$ leads to the final state: $\psi_1\rangle = e^{i\frac{\phi}{2}} \left(\cos \frac{\phi}{2} 0\rangle - i \sin \frac{\phi}{2} 1\rangle \right)$ and measurement of the qubit leads to an interference fringe: $P(0) = \cos^2 \frac{\phi}{2}$ and $P(1) = \sin^2 \frac{\phi}{2}$.
(P) Step 2: coupling kills interference	Coupling the first qubit with another qubit kills the interference as the setup: $\begin{array}{c} 0\rangle \text{---} \boxed{H} \text{---} \bullet \text{---} \boxed{\phi} \text{---} \boxed{H} \text{---} \\ 0\rangle \text{---} \oplus \text{---} \end{array}$ leads to the final state: $\psi_2\rangle = \frac{1}{2} \left(00\rangle + 10\rangle + e^{i\phi} 01\rangle - e^{i\phi} 11\rangle \right)$ and measurement of the first qubit gives a result without the interference pattern: $P(0) = \frac{1}{2}$ and $P(1) = \frac{1}{2}$, actually all $P(ij)$ are $\frac{1}{4}$
(P) Step 3: quantum “eraser”	Now if you “erase” the information from the second qubit by measuring it in a conjugate basis, then you do get back the interference from the first qubit, only if you consider the results conditional to the measurement outcome of the second qubit. The following setup: $\begin{array}{c} 0\rangle \text{---} \boxed{H} \text{---} \bullet \text{---} \boxed{\phi} \text{---} \boxed{H} \text{---} \\ 0\rangle \text{---} \oplus \text{---} \boxed{H} \text{---} \end{array}$ leads to the final state: $\psi_3\rangle = \frac{e^{i\frac{\phi}{2}}}{\sqrt{2}} \left(\cos \frac{\phi}{2} 00\rangle - i \sin \frac{\phi}{2} 10\rangle - i \sin \frac{\phi}{2} 01\rangle + \cos \frac{\phi}{2} 11\rangle \right)$ which does not change the marginal probability for the measurement of qubit 1 which stays at $\frac{1}{2}$ (otherwise you could have superluminal signaling between the qubit 1 and 2), but if you consider results from qubit 1, conditional on qubit 2 giving 0 for instance, you have interference pattern: $P(0 0) = \frac{P(00)}{P(00)+P(10)} = \cos^2 \frac{\phi}{2}$, and $P(1 0) = \sin^2 \frac{\phi}{2}$
(P) Step 4: delayed quantum ‘eraser’	You can also “delay” whether the change of basis for qubit 2 had been performed or not by coupling it to a third qubit: $\begin{array}{c} 0\rangle \text{---} \boxed{H} \text{---} \bullet \text{---} \boxed{\phi} \text{---} \boxed{H} \text{---} \\ 0\rangle \text{---} \oplus \text{---} \boxed{H} \text{---} \\ 0\rangle \text{---} \boxed{H} \text{---} \bullet \text{---} \end{array}$ leading to the final state $\psi_4\rangle = \psi_2\rangle \otimes 0\rangle + \psi_3\rangle \otimes 1\rangle$.

cloning and U-NOT gate	
(P) approximative cloning and U-NOT gate	Cloning a quantum state ($\phi\rangle \otimes 0\rangle \mapsto \phi\rangle \otimes \phi\rangle$) is not possible. Performing a Universal-Not (for a qubit $\phi\rangle = \alpha 0\rangle + \beta 1\rangle$, a U-NOT maps $\phi\rangle$ to $\phi_\perp\rangle = \beta^* 0\rangle - \alpha^* 1\rangle$) is not possible because it is anti-unitary (A is anti-unitary if for any $x\rangle$ and $y\rangle$, $\langle Ax Ay\rangle = (\langle x y\rangle)^* = \langle y x\rangle$) However, the following circuit performs an approximate cloning and U-NOT  where $\chi\rangle = \frac{\mu}{\sqrt{2}}(00\rangle + 11\rangle) + \frac{\nu}{\sqrt{2}}(00\rangle + 01\rangle)$, where μ and ν are real numbers, in the sense that $\rho_1 = \text{Tr}_{23}(\xi\rangle\langle\xi) = (\mu^2 + \mu\nu) \psi\rangle\langle\psi + \frac{\nu^2}{2}\mathbf{1}$ $\rho_2 = \text{Tr}_{13}(\xi\rangle\langle\xi) = (\nu^2 + \mu\nu) \psi\rangle\langle\psi + \frac{\mu^2}{2}\mathbf{1}$ $\rho_3 = \text{Tr}_{12}(\xi\rangle\langle\xi) = \frac{1}{2}(1 + \mu\nu) \psi_\perp\rangle\langle\psi_\perp + \frac{1}{2}(1 - \mu\nu) \psi\rangle\langle\psi .$ For $\mu = \nu = 1/\sqrt{3}$ one obtains $\rho_1 = \rho_2 = \frac{5}{6} \psi\rangle\langle\psi + \frac{1}{6} \psi_\perp\rangle\langle\psi_\perp $ i.e. qubits 1 and 2 are clone of $\psi\rangle$ with fidelity 5/6, and $\rho_3 = \frac{2}{3} \psi_\perp\rangle\langle\psi_\perp + \frac{1}{3} \psi\rangle\langle\psi .$

programmable machines	
(D) programmable machine G	A programmable machine is a unitary operator G acting on a product space $H_{data} \otimes H_{program}$, where the program state $\chi\rangle \in H_{program}$ specifies what operation is to be applied on the data state $\psi\rangle \in H_{data}$.
(P) Nielsen and Chuang	Each unitary operator that the machine can implement requires an additional dimension in the program space $H_{program}$. More precisely, if $\chi_1\rangle \in H_{program}$ implements the unitary operator $U_1 \in \mathcal{L}(H_{data})$ and $\chi_2\rangle \in H_{program}$ implements the unitary operator $U_2 \in \mathcal{L}(H_{data})$, then: $\langle\chi_1 \chi_2\rangle = 0$, or there exists $\theta \in R$ such that $U_1 = e^{i\theta}U_2$.
(Proof) For any $\psi\rangle \in H_{data}$, there exists $\xi(\psi)\rangle \in H_{program}$ that could depend on $\psi\rangle$ so that $G \psi\rangle \otimes \chi\rangle = U \psi\rangle \otimes \xi(\psi)\rangle$. So for any two states $\psi\rangle$ and $\phi\rangle \in H_{data}$, we have $\langle\phi \otimes \langle\chi G^\dagger G \psi\rangle \otimes \chi\rangle = \langle\phi \psi\rangle = \langle\phi U^\dagger U \psi\rangle \langle\xi(\phi) \xi(\psi)\rangle$, which proves that $\langle\xi(\phi) \xi(\psi)\rangle = 1$, and $\xi(\psi)\rangle = \xi\rangle$ actually does not depend on the state $\psi\rangle$. Now if you have $\chi_1\rangle$ implementing U_1 and $\chi_2\rangle$ implementing U_2, you have $G \psi\rangle \otimes \chi_1\rangle = U \psi\rangle \otimes \xi_1\rangle$, and $G \psi\rangle \otimes \chi_2\rangle = U \psi\rangle \otimes \xi_2\rangle$, and taking the inner product we get $\langle\chi_1 \chi_2\rangle = \langle\psi U_2^\dagger U_1 \psi\rangle \langle\xi_1 \xi_2\rangle$. If $\langle\xi_1 \xi_2\rangle$ is not zero, then we have $\langle\psi U_2^\dagger U_1 \psi\rangle = \langle\chi_1 \chi_2\rangle / \langle\xi_1 \xi_2\rangle$ with the rhs being independent of $\psi\rangle$ which means $U_2^\dagger = U_1^{-1}$ modulo a phase factor. The other alternative is $\langle\xi_1 \xi_2\rangle$ to be zero which implies $\langle\chi_1 \chi_2\rangle$ to be zero as well.	

Kochen-Specker paradox	
(D) Spin-1 matrices	$S_1 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -i \\ 0 & i & 0 \end{pmatrix} \quad S_2 = \begin{pmatrix} 0 & 0 & i \\ 0 & 0 & 0 \\ -i & 0 & 0 \end{pmatrix} \quad S_3 = \begin{pmatrix} 0 & -i & 0 \\ i & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$ For any vector $\vec{u} = (u_1, u_2, u_3) \in R^3$, $S_{\vec{u}} = \sum_j u_j S_j$
(P) Properties	(1) $[S_1, S_2] = iS_3$, $[S_2, S_3] = iS_1$, $[S_3, S_1] = iS_2$ (2) $[S_i^2, S_j^2] = 0$, for all i, j (3) S_i^2, has eigenvalues 0 (rank 1) and 1 (rank 2), for all i (4) $S_1^2 + S_2^2 + S_3^2 = 2 \times \mathbf{1}$
(P) 1-0-1 rule	For any orthonormal basis $\{\vec{a}, \vec{b}, \vec{c}\}$ of R^3 , the operators $S_{\vec{a}}^2, S_{\vec{b}}^2$ and $S_{\vec{c}}^2$ commute and their measurement yields the outcomes 1, 0, 1 in some order.
(P) Kochen-Specker paradox	There exists no deterministic mapping $f : \vec{u} \in R^3 \mapsto f(\vec{u}) \in \{0, 1\}$ that reproduces quantum mechanics prediction for $S_{\vec{u}}^2$ for any $\vec{u} \in R^3$. This result is usually referred to as another theorem (with Bell's inequality) going against hidden variable theories.
(Proof) see for instance The Strong Free Will Theorem by Conway and Kochen, Notice of the AMS, Vol 56, Number 2 p 226	